Michigan Tech

Circular Restricted Three-Body Problem

Derivation in rotating coordinates

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Math Sciences

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Outline

1. Add a massless probe to the two-body system.
2. Choose circular motion for the two primaries.
3. Transform the probe acceleration to rotating coordinates.
4. Obtain the coupled equations for $x(t)$ and $y(t)$.

Three-body acceleration equations

- The **Primaries** have masses $m_1, m_2 > 0$. The probe has $m_3 \approx 0$.

$$\boldsymbol{r}_i'' = -G \sum_{\substack{j=1 \\ j \neq i}}^{3} m_j \frac{\boldsymbol{r}_i - \boldsymbol{r}_j}{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^3} \qquad i = 1, 2, 3$$

- If if $m_3/m_1 \approx 0$ and $m_3/m_2 \approx 0$ the probes effect on the primaries is negligible. Letting $m_3 \to 0$ in the acceleration equations gives the restricted problem [1].

Massless probe limit

- Terms proportional to m_3 disappear from the primary equations.

$$\boldsymbol{r}_1'' = -Gm_2 \frac{\boldsymbol{r}_1 - \boldsymbol{r}_2}{\|\boldsymbol{r}_1 - \boldsymbol{r}_2\|^3}$$
$$\boldsymbol{r}_2'' = -Gm_1 \frac{\boldsymbol{r}_2 - \boldsymbol{r}_1}{\|\boldsymbol{r}_2 - \boldsymbol{r}_1\|^3}$$

- The probe still accelerates under both gravitational fields.

$$\boldsymbol{r}_3'' = -Gm_1 \frac{\boldsymbol{r}_3 - \boldsymbol{r}_1}{\|\boldsymbol{r}_3 - \boldsymbol{r}_1\|^3} - Gm_2 \frac{\boldsymbol{r}_3 - \boldsymbol{r}_2}{\|\boldsymbol{r}_3 - \boldsymbol{r}_2\|^3}$$

- The probe has no effect on the primaries.

Circular primary motion

1. Recall the relative displacement equation with $M = m_1 + m_2$.

$$\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1 \qquad \boldsymbol{r}'' = -GM\frac{\boldsymbol{r}}{\|\boldsymbol{r}\|^3}$$

2. Choose the center-of-mass rest frame, $\boldsymbol{c} = \boldsymbol{0}$ and the circular $\epsilon = 0$ orbit.

$$\boldsymbol{r} = a(\cos\omega t, \sin\omega t) \qquad \boldsymbol{r}'' = -\omega^2\boldsymbol{r}$$

3. For constant separation $a > 0$, comparison gives

$$\boxed{\omega^2 = \frac{GM}{a^3}} \qquad \omega > 0$$

Primary positions about the center of mass

- Define the **Mass Fraction**

$$\eta = \frac{m_2}{M} \qquad 1 - \eta = \frac{m_1}{M} \qquad 0 < \eta < 1$$

- The reconstruction formulas from the two-body deck give

$$\boldsymbol{r}_1 = -\eta a(\cos\omega t, \sin\omega t)$$
$$\boldsymbol{r}_2 = (1-\eta) a(\cos\omega t, \sin\omega t)$$

- The mass fraction η and reduced mass $\mu = m_1 m_2 / M$ are dimensionless.

Planar restricted problem

- Circular primaries remain in a fixed plane.
- Probe's initial position and initial velocity are in plane. The probe remains planar until a collision.
- This is the **Planar Circular Restricted Three-Body Problem** (PCR3BP) [1].
- Circular refers to the primary orbits. The probe's orbit is not circular.

Rotating coordinates

- Rotate the axes counterclockwise at the primary angular speed ω.

$$\boldsymbol{R}(t) = \begin{pmatrix} \cos\omega t & -\sin\omega t \\ \sin\omega t & \cos\omega t \end{pmatrix}$$

- Write the probe's rotating position as $\boldsymbol{q} = (x, y)$.

$$\boldsymbol{r}_3 = \boldsymbol{R}(t)\boldsymbol{q}$$

- The primary coordinates are fixed.

$$\boldsymbol{q}_1 = (-\eta a, 0) \qquad \boldsymbol{q}_2 = ((1-\eta)a, 0)$$

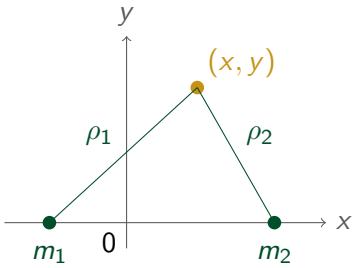
Probe distances in rotating coordinates

- Define the distances to the primaries.

$$\rho_1 = \sqrt{(x+\eta a)^2 + y^2}$$

$$\rho_2 = \sqrt{(x-(1-\eta)a)^2 + y^2}$$

- Rotation preserves lengths, so $\rho_i = \|\boldsymbol{r}_3 - \boldsymbol{r}_i\|$.
- The equations exclude $\rho_1 = 0$ and $\rho_2 = 0$.



Schematic

Differentiating the rotation matrix

- Introduce the constant quarter-turn matrix.

$$\boldsymbol{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \qquad \boldsymbol{J}^2 = -\boldsymbol{I} \qquad \boldsymbol{J}\boldsymbol{q} = (-y, x)$$

- Differentiate $\boldsymbol{R}(t)$.

$$\boldsymbol{R}' = \omega \boldsymbol{R}\boldsymbol{J} \qquad \boldsymbol{R}'' = -\omega^2 \boldsymbol{R}$$

- ω is constant because the primary orbits are circular.

Probe velocity transformation

1. Differentiate $\boldsymbol{r}_3 = \boldsymbol{R}\boldsymbol{q}$ using the product rule.

$$\boldsymbol{r}_3' = \boldsymbol{R}'\boldsymbol{q} + \boldsymbol{R}\boldsymbol{q}'$$

2. Substitute $\boldsymbol{R}' = \omega \boldsymbol{R}\boldsymbol{J}$.

$$\boxed{\boldsymbol{r}_3' = \boldsymbol{R}(\boldsymbol{q}' + \omega \boldsymbol{J}\boldsymbol{q})}$$

3. The inertial velocity combines motion relative to the axes with motion from rotation of the axes.

Probe acceleration transformation

1. Differentiate the position twice.

$$\boldsymbol{r}_3'' = \boldsymbol{R}\boldsymbol{q}'' + 2\boldsymbol{R}'\boldsymbol{q}' + \boldsymbol{R}''\boldsymbol{q}$$

2. Substitute the rotation derivatives.

$$\boxed{\boldsymbol{r}_3'' = \boldsymbol{R}(\boldsymbol{q}'' + 2\omega\boldsymbol{J}\boldsymbol{q}' - \omega^2\boldsymbol{q})}$$

3. Thus the inertial acceleration expressed along the rotating axes has components

$$\left(x'' - 2\omega y' - \omega^2 x,\ y'' + 2\omega x' - \omega^2 y\right)$$

Gravity transformation

1. Each relative displacement transforms by the same rotation.

$$\boldsymbol{r}_3 - \boldsymbol{r}_i = \boldsymbol{R}(\boldsymbol{q} - \boldsymbol{q}_i) \qquad i = 1, 2$$

2. Substitute into the probe's inertial acceleration equation.

$$\boldsymbol{R}(\boldsymbol{q}'' + 2\omega \boldsymbol{J}\boldsymbol{q}' - \omega^2 \boldsymbol{q}) = \boldsymbol{R}\left[-G\sum_{i=1}^{2} m_i \frac{\boldsymbol{q} - \boldsymbol{q}_i}{\rho_i^3}\right]$$

3. Multiply by $\boldsymbol{R}^{-1}$ to obtain the rotating equation.

$$\boldsymbol{q}'' + 2\omega \boldsymbol{J}\boldsymbol{q}' = \omega^2 \boldsymbol{q} - G\sum_{i=1}^{2} m_i \frac{\boldsymbol{q} - \boldsymbol{q}_i}{\rho_i^3}$$

Rotation terms and gravitational coupling

- Move the velocity term to the right.

$$\boldsymbol{q}'' = -2\omega \boldsymbol{J}\boldsymbol{q}' + \omega^2 \boldsymbol{q} - G\sum_{i=1}^{2} m_i \frac{\boldsymbol{q} - \boldsymbol{q}_i}{\rho_i^3}$$

- The **Coriolis Acceleration** is $-2\omega \boldsymbol{J}\boldsymbol{q}' = (2\omega y', -2\omega x')$.
- The **Centrifugal Acceleration** is $\omega^2 \boldsymbol{q} = (\omega^2 x, \omega^2 y)$ [2].
- Both distances ρ_1, ρ_2 depend on x and y, so gravity also couples the coordinates.

Dimensional coupled equations

- Substitute the fixed primary positions and equate components [2].

$$x'' - 2\omega y' = \omega^2 x - Gm_1 \frac{x + \eta a}{\rho_1^3} - Gm_2 \frac{x - (1-\eta)a}{\rho_2^3}$$

$$y'' + 2\omega x' = \omega^2 y - Gm_1 \frac{y}{\rho_1^3} - Gm_2 \frac{y}{\rho_2^3}$$

- The parameters and distances are

$$\eta = \frac{m_2}{m_1 + m_2} \qquad \omega^2 = \frac{G(m_1 + m_2)}{a^3}$$

$$\rho_1^2 = (x + \eta a)^2 + y^2 \qquad \rho_2^2 = (x - (1-\eta)a)^2 + y^2$$

Standard scaled coordinates

- Use primary separation a as the length unit and $1/\omega$ as the time unit.

$$x = \frac{x_{\text{phys}}}{a} \qquad y = \frac{y_{\text{phys}}}{a} \qquad t = \omega t_{\text{phys}}$$

- From this slide onward, x, y, t and ρ_i are dimensionless. Primes denote derivatives with respect to scaled t.

$$\frac{d^2 x_{\text{phys}}}{dt_{\text{phys}}^2} = a\omega^2 x'' \qquad \frac{dy_{\text{phys}}}{dt_{\text{phys}}} = a\omega y'$$

- Divide each dimensional equation by $a\omega^2$.

$$\frac{Gm_1}{a^3\omega^2} = 1 - \eta \qquad \frac{Gm_2}{a^3\omega^2} = \eta$$

Coupled equations in rotating coordinates

- The scaled planar circular restricted three-body system is

$$\boxed{\begin{aligned} x'' - 2y' &= x - (1-\eta)\frac{x+\eta}{\rho_1^3} - \eta\frac{x-1+\eta}{\rho_2^3} \\ y'' + 2x' &= y - (1-\eta)\frac{y}{\rho_1^3} - \eta\frac{y}{\rho_2^3} \end{aligned}}$$

- The probe is at (x, y), and the primaries are at $(-\eta, 0)$ and $(1-\eta, 0)$.

$$\rho_1 = \sqrt{(x+\eta)^2 + y^2} \qquad \rho_2 = \sqrt{(x-1+\eta)^2 + y^2}$$

- $\eta = m_2/(m_1 + m_2)$ is the only mass parameter; $\rho_1, \rho_2 > 0$ [2].

References I

[1] Richard Fitzpatrick.
Circular restricted three-body problem.
Introduction to Celestial Mechanics, 2nd edition, University of Texas at Austin.
Section 9.2; accessed 12 October 2026.

[2] Richard Fitzpatrick.
Co-rotating frame.
Introduction to Celestial Mechanics, 2nd edition, University of Texas at Austin.
Section 9.5; accessed 12 October 2026.